# Linear Combination and Coplanar Vectors

*Recall*: Two vectors $\vec{u}$ and $\vec{v}$ are collinear if $\vec{u} = k\vec{v}$, $k \in \Re$. (i.e. They are scalar multiples).

## Linear Combination in $R^2$

Linear combination of vectors is when we express one vector as a sum of two other vectors.

i.e. $\overrightarrow{OP} = (3,2)$ can be expressed as $\overrightarrow{OP} = 3\hat{i} + 2\hat{j}$

Every vector in $R^2$ can be written as a linear combination of $\hat{i}$ and $\hat{j}$, therefore $\hat{i}$ and $\hat{j}$ are said to FORM A BASIS for $R^2$. In fact, they are called the STANDARD BASIS vectors for $R^2$.

In fact, as long as two vectors $\vec{u}$ and $\vec{v}$ are NOT COLLINEAR, they can also form a basis in $R^2$.

**Example 1:** Express $\vec{w} = (6,8)$ as a linear combination of $\vec{u} = (3,5)$ and $\vec{v} = (1,2)$.

Let $a, b \in R$ $\quad \vec{w} = a\vec{u} + b\vec{v}$

$(6,8) = a(3,5) + b(1,2)$

$\therefore 6 = 3a + b$ ①

$8 = 5a + 2b$ ②

use sub/ elimination

① × 2 $\quad 12 = 6a + 2b$

− ② $\quad 8 = 5a + 2b$

$4 = a$

Sub $a = 4$ into ①

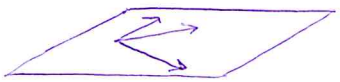
$6 = 3(4) + b$

$6 = 12 + b$

$-6 = b$

$\therefore \vec{w} = 4\vec{u} - 6\vec{v}$

## Coplanar Vectors in $R^3$



Vectors are said to be COPLANAR if they lie in the same plane. (i.e. one can be expressed as a linear combination of the other two).

Three vectors, $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar if and only if there exists two scalars a and b such that $\vec{w} = a\vec{u} + b\vec{v}$.

**Example 2:** Determine if $\vec{u} = (3,-1,4)$, $\vec{v} = (6,-4,-8)$, and $\vec{w} = (7,-3,4)$ are coplanar?

only if $\vec{w} = a\vec{u} + b\vec{v}$ , $a, b \in \mathbb{R}$.

$(7,-3,4) = a(3,-1,4) + b(6,-4,-8)$

$\therefore$ $7 = 3a + 6b$ ①
$-3 = -a - 4b$ ②
$4 = 4a - 8b$ ③

① $7 = 3a + 6b$
$+$ ②$\times 3$ $-9 = -3a - 12b$

$-2 = -6b$

$\frac{1}{3} = b$

Sub $b = \frac{1}{3}$ into ①

$7 = 3a + 6(\frac{1}{3})$

$7 = 3a + 2$

$5 = 3a$

$\frac{5}{3} = a$

$\therefore$ Yes! Coplanar. $\vec{w} = \frac{5}{3}\vec{u} + \frac{1}{3}\vec{v}$

* Need to check in ③ *. to make sure true for all components.

LS $= 4$

RS $= 4(\frac{5}{3}) - 8(\frac{1}{3})$
$= \frac{20}{3} - \frac{8}{3}$
$= \frac{12}{3}$
$= 4$

$\therefore$ LS = RS

Is there a quicker way to determine if 3 vectors are coplanar?

**Triple Scalar Product** $\vec{u} \bullet (\vec{v} \times \vec{w})$

Recall: $|\vec{u} \cdot (\vec{v} \times \vec{w})|$ = Volume of parallelopiped.

If $\vec{u} \bullet (\vec{v} \times \vec{w}) = 0$, then the non-zero vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar.

if volume = 0 coplanar !!

**Example 3:** Determine if $\vec{u} = (3,-1,4)$, $\vec{v} = (6,-4,-8)$, and $\vec{w} = (7,-3,4)$ are coplanar?

$\vec{u} \cdot \vec{v} \times \vec{w} = (3,-1,4) \cdot$ $\begin{matrix} 6 & -4 & -8 & 6 & -4 \\ 7 & -3 & 4 & 7 & -3 \end{matrix}$ $\times$

$= (3,-1,4) \cdot (-40,-80,10)$

$= -120 + 80 + 40$

$= 0$

$\therefore$ $\vec{u}, \vec{v} \& \vec{w}$ are coplanar.